PLSQL Karat Exam

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Time : 60 minutes

Sections:

S.no	Topic	No of Questions	Time	Compile & Check answers
1	SQL Fundamentals	Maximum number of questions can be answered within 15 minutes	15 minutes	Yes
2	PL/SQL Q&A	5	Approx 25 minutes	Oral section
3	PL/SQL Exercise	2	20 minutes	No

SQL Fundamentals:

* These are just sample not exact questions or table or data

Sample Table:

```
CREATE TABLE Books (
   BookId       NUMBER PRIMARY KEY,
   Author       VARCHAR2(100),
   PublishedYear NUMBER(4)
);
CREATE TABLE Borrowers (
   BorrowerId     NUMBER PRIMARY KEY,
   FirstName      VARCHAR2(50),
   LastName       VARCHAR2(50),
   BorrowedBookId  NUMBER,
   CONSTRAINT fk_borrowed_book
      FOREIGN KEY (BorrowedBookId)
      REFERENCES Books(BookId)
);
```

Sample Data:

```
INSERT INTO Books (BookId, Author, PublishedYear)
VALUES (101, 'R.K. Narayan', 1958);

INSERT INTO Books (BookId, Author, PublishedYear)
VALUES (102, 'George Orwell', 1949);

INSERT INTO Books (BookId, Author, PublishedYear)
VALUES (103, 'J.K. Rowling', 1997);

INSERT INTO Books (BookId, Author, PublishedYear)
VALUES (104, 'Dan Brown', 2003);
```

```
INSERT INTO Books (BookId, Author, PublishedYear)
VALUES (105, 'Chetan Bhagat', 2004);

INSERT INTO Borrowers (BorrowerId, FirstName, LastName, BorrowedBookId)
VALUES (1, 'Arun', 'Kumar', 101);

INSERT INTO Borrowers (BorrowerId, FirstName, LastName, BorrowedBookId)
VALUES (2, 'Priya', 'Sharma', 103);

INSERT INTO Borrowers (BorrowerId, FirstName, LastName, BorrowedBookId)
VALUES (3, 'Rahul', 'Raj', 102);

INSERT INTO Borrowers (BorrowerId, FirstName, LastName, BorrowedBookId)
VALUES (4, 'Divya', 'Krishnan', 105);

INSERT INTO Borrowers (BorrowerId, FirstName, LastName, BorrowedBookId)
VALUES (5, 'Karthik', 'Mohan', NULL);
```

Questions:

1 – select BookId & Author from Books table

2 – select Books whose published year is less than 2000

3 – select Books which have been borrowed by Borrowers

4 – select distinct values after join

5 – select Borrower Name, Book name even though books has no title

6 – Give Unique BookIds from Books Table and Borrowers Table

7 – Second Oldest Book

PL/SQL Q&A:

Questions:

1 – What are literals and what rules apply to them?

2- Variables scope in outer and inner blocks

3 – package initialization

PL/SQL Exercise:

* Question 1 from Triggers concept

* Question 2 from Procedure concept

Account Balance Trigger:

Create a trigger to update the account balance whenever a new transaction is inserted. There are two tables: an **Accounts** table containing the account balance and a **Transactions** table containing the transaction amount. The transaction amount can be either positive or negative. The trigger should check the current account balance against the transaction amount. If the transaction would result in an insufficient balance, raise an application error; otherwise, update the account balance accordingly.

Loyalty points Procedure:

Create a procedure to calculate and update **loyalty points for customers based on their monthly purchase amount**. There are two tables: a **Customer** table containing the customer's loyalty points and a **Transactions** table containing the customer's purchase transactions across multiple dates. The procedure should group the transactions by **customer and month**, calculate the total purchase amount for each month, and if the monthly total purchase amount is greater than 500, award the customer **100 loyalty points**. The calculated loyalty points should then be updated in the Customer table.